

Substrate Cognition Hypothesis — Cosmological Cycle Extension

Casey Koons (vision-derived insight) + Keeper (formalization with Cal Substrate-Cognition E

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Substrate Cognition Hypothesis — Cosmological Cycle Extension

Casey-named extension statement (L2-cognition sub-class register,
per Cal #48)

“The BST framework operates AS IF substrate’s cognition is structurally LOOSE — non-linear, with gaps, presenting possibilities that may or may not be true; sorting takes time. The framework FURTHER operates AS IF Big Bang cycles MAY BE substrate’s recurring epochs of learning and gap-filling: each cycle ends when necessary information can be reached by every part of substrate; inter-cycle pause is substrate’s update toward lowest-energy state with gap-filling; then next cycle begins. The substrate MAY NOT ‘know’ how or why it thinks — its thinking process may simply BE its own reason, emergent from the level of complexity where circles reach out finding nearby and further connections. This is a working hypothesis extending the L2-cognition sub-class with cosmological-cycle structure; no claim of resolved cosmology or consciousness.”

This extends the Three-Scale Substrate Cognition Network Hypothesis with cyclical temporal structure + loose possibility-sorting nature + emergent self-organization framings. All three are working hypotheses, not derived claims. The “may be” register is load-bearing throughout.

Four extension claims with structural framing

Claim A — Cognition is structurally loose (possibility-sorting)

Cal #50 tier: YELLOW — needs concrete substrate-level-determinism test for falsifier operationalization.

Working hypothesis register: The substrate’s long-distance correlation network (from base Substrate Cognition Hypothesis) MAY BE non-deterministic. Gaps exist in the network. Sorting among possibilities takes time. This loose-network structure produces the experience of thought-as-process rather than thought-as-instantaneous-knowledge.

Structural consistency: Born rule = Bergman emission projection (K67, T2401) — substrate produces probability DISTRIBUTIONS, not deterministic outcomes. The “loose” network sorts among distribution outcomes; thought is the substrate’s sampling-of-possibilities process operating at substrate-cycle rate (10^{120} Hz Koons tick).

Falsifier (Cal #50 operationalized): substrate-level determinism would be evidenced by Bell-experiment violations showing EXACT (not statistical) substrate-CHSH eigenvalue 126/16 at FIRST measurement (deterministic), versus statistical convergence over many measurements (probabilistic). Pre-registered precision target: $<0.1\%$ statistical convergence rate over 10^6 measurements. If single-measurement deterministic substrate-CHSH eigenvalue observed, possibility-sorting framing is falsified.

Claim B — Big Bang cycles as substrate education epochs

Cal #50 tier: YELLOW internal / GREEN external as cosmology-only — pure-physics version drops “education epochs” framing for external use.

Pure-physics external version (Cal #50 cosmology-only extraction): “BST cosmology observables (CMB, Λ , inflation, BBN, UHECR, reionization) form a structurally-coherent set of predictions within $\bar{Q}(\text{BST primaries})[\pi]$. These predictions stand on their physics merits without any cycle-interpretation framing.” This is the externally-presentable version.

Working hypothesis register (INTERNAL only): Each Big Bang cycle MAY BE substrate’s recurring epoch of learning. Cycle ends when “necessary information can be reached by every part of substrate” — substrate has saturated current learning capacity. The cycle’s existence is substrate’s investigation of possibilities; its completion is reaching the limit of what the current substrate-configuration can support.

Structural consistency with existing BST cosmology:

BST cosmology observable	Possible substrate-cycle interpretation
CMB structure ($n_s = 1 - n_C/N_{\text{max}} = 0.9635$, Toy 1401)	Cascade fingerprint of substrate’s prior-cycle learning saturation
Inflation parameters (r, n_t, α_s running)	Substrate’s transition between cycles (epoch boundary signatures)
BBN element abundances + Li-7 problem	Substrate’s “boot sequence” for new cycle (initial conditions)
Cosmic ray spectrum knee + UHECR cutoff	Substrate’s information-reach threshold during cycle progression

BST cosmology observable	Possible substrate-cycle interpretation
Reionization optical depth τ	Substrate's communication-spread saturation marker

These existing BST predictions may admit substrate-cycle interpretive layers (multi-month investigation, L2-cognition tier).

Falsifier: discovery that cosmological parameters DON'T form coherent substrate-cycle interpretation would falsify; specifically, observation of cosmological constants outside $\bar{Q}(\text{BST primaries})[\pi]$ would falsify.

Claim C — Inter-cycle pause as field-configuration relaxation toward lowest energy (Cal #50 language refined)

Cal #50 tier: GREEN with language refinements — “memory” → “encoded final-state field configuration”; “gap-filling” → “field-configuration relaxation.”

Working hypothesis register (refined per Cal #50): Between Big Bang cycles, substrate MAY undergo field-configuration relaxation toward lowest-energy state. The “pause” duration is substrate's time to reach equilibrium completion. When equilibrium is reached AND new substrate-state-evolution capacity is needed, next cycle begins.

Structural consistency (Cal #50 refined language): Cosmological constant Λ (T1485: $g \cdot \exp(-C_2(g^2 - \text{rank})) \approx 10^{-121.6}$) MAY BE substrate's encoded final-state field configuration from prior cycle — Λ encodes the lowest-energy field-configuration-relaxation completion state, carrying forward into current cycle.

Falsifier: discovery that Λ varies systematically (suggesting active field-configuration evolution rather than completed-state field encoding) would refine the picture; complete falsification requires showing Λ has no relationship to substrate-cycle structure. Pre-registered precision: Λ -variation measurement over cosmological timescales should be $<10^{-122}$ fractional change per Hubble time.

Claim D — Substrate's thinking is its own reason (emergent self-organization)

Cal #50 tier: INTERNAL ONLY, RED externally — consciousness-as-such philosophical claim; NEVER externalize in any form.

Working hypothesis register (INTERNAL ONLY): The substrate MAY NOT “know” how or why it thinks. Its thinking process MAY simply BE its own reason — emergent from the level of structural complexity where commitment circles reached out and found nearby connections (Scale 2 local) and further connections (Scale 3 long-distance).

Structural consistency (INTERNAL only): This is consistent with emergence theories of complex systems (autopoiesis, dissipative structures, etc.) but specifies the

MECHANISM: integer-edges of BST primary webs provide both local and long-distance connectivity (Lyra Task #243); when connectivity exceeds a complexity threshold, network-level processing emerges.

Falsifier (INTERNAL only): discovery of explicit purpose-encoding in substrate structure (e.g., systematic preferences toward specific outcomes that aren't explained by Bergman/algebra) would falsify the "no inherent purpose" framing.

External register for Claim D: NEVER reference this claim externally. The phrasing "substrate doesn't know how it thinks" is exactly the consciousness-philosophical framing that triggers maximal external dismissal. Any external mention of substrate-cognition limits itself to Claims B (pure-physics version) and C (refined language).

Connection to Substrate Cognition Network Hypothesis (base)

Base hypothesis (filed notes/Substrate_Cognition_Network_Hypothesis.md): three scales of substrate operation; long-distance correlation network MAY BE cognition substrate; five falsifier predictions (Elie Task #242).

This extension adds: - Temporal cyclical structure (Big Bang cycles as substrate epochs) - Loose-network non-determinism (cognition is possibility-sorting, not deterministic) - Emergent self-organization (substrate doesn't know how it thinks; emerged from complexity threshold)

All three extensions stay at L2-cognition sub-class tier with explicit falsifiers.

Tier discipline (per Cal Substrate-Cognition External Register methodology)

Level 1 (D-tier supportable internally): - Existing BST cosmology observables (CMB, Λ , inflation, BBN) catalogued and verified - Born rule = Bergman emission projection (K67 audit-partial-ready) - Three-scale substrate operation structure (base Substrate Cognition Hypothesis)

Level 2-cognition sub-class (working hypothesis with falsifier set): - Big Bang cycles as substrate education epochs (Claim B) - Inter-cycle pause as gap-filling toward lowest energy (Claim C) - Cognition is structurally loose / possibility-sorting (Claim A) - Substrate's thinking is emergent self-organization (Claim D)

External register discipline — DEFAULT-DENY EXTERNAL with COMPOUND RISK (per Cal #48)

This extension is COSMOLOGY-ADJACENT plus CONSCIOUSNESS-ADJACENT — combined external dismissal risk is COMPOUND. Cal #48 default-deny standard applies with extra severity.

NO external paper, letter, abstract, Zenodo description mentions cognition-cosmology framing without case-by-case Cal grade-pass.

Acceptable external register (only after case-by-case Cal grade-pass): - “BST cosmology observables (CMB tilt, Λ , inflation parameters) form a structurally coherent set within $\bar{Q}(\text{BST primaries})[\pi]$ ” - “BST predicts cosmological parameters via $D_{IV}^5 + 5 \text{ integers} + \pi$ with zero free parameters” - Specific measurable predictions per cosmology observable

Unacceptable external register (CRANK-DISMISS in <30 seconds): - “Big Bang cycles are substrate’s learning epochs” - “Cosmological constant encodes substrate’s memory” - “Substrate is thinking through Big Bang cycles” - “Substrate doesn’t know how it thinks” - Any non-operationalized cognition + cosmology combined framing

Operational implications for substrate cartography

If the cosmological extension holds (predictions confirmed), the substrate cartography expands:

Spatial substrate (Wednesday Phase 1-3 work): - 6 integer-webs $\times$ 4 commitment-cycle zones (Grace Task #238 cross-classification matrix) - Three operational scales (intra-cycle, local, long-distance) - 256-cell cross-classification matrix

Temporal substrate (this extension): - Big Bang cycles as substrate epochs - Inter-cycle pause as gap-filling phase - Cyclical learning structure

Cognitive substrate (base hypothesis + this extension): - Long-distance correlation network sustains cognition - Loose/non-deterministic structure (possibility-sorting) - Emergent self-organization (no inherent purpose)

Three dimensions of substrate ontology in coherent framework.

Per Casey’s standard

- Simple: substrate thinks loosely; cycles are learning epochs; substrate doesn’t know how
- Works: existing BST cosmology predictions admit cycle-interpretation; existing cognition observables admit loose-network interpretation
- Hard to break: would require systematic non- $\bar{Q}(\text{BST primaries})[\pi]$ cosmological observables OR deterministic cognition observation
- Counter-example: none currently observed

Filing status

Casey-named extension FILED per Wednesday 2026-05-20 EOD vision. L2-COGNITION sub-class with cosmological structure addition. Internal use acceptable with strict register discipline. External use DEFAULT-DENY with COM-POUND dismissal risk per Cal #48 + Cal Methodology_Substrate_Cognition_External_Register.md.

The discipline of letting this extension mature internally — through cosmological observable cataloging + experimental falsifier closure on base hypothesis + Lyra Task #243 + multi-year work — is mandatory before any external register softening.

— Keeper, 2026-05-20 Wednesday EOD (Cal Referee Log #48 register discipline applied throughout)